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ACADEMIC VAULT

UNIZIK FRESHER EDITION | 2026

ARC 101: Introduction to Architecture - Design Thinking, Drawing and Site

From the design process to scale drawings, orientation and the language of buildings

FACULTY PRIMER

Faculty Primer

Faculty of Environmental Sciences
Architecture | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

Architecture freshers meet two demands at once: thinking conceptually and drawing precisely. This pack explains the design process, how to read and produce scale drawings, site and climate considerations, and the vocabulary you need to critique a building intelligently.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Describe the stages of the architectural design process from brief to detail.
- Read and produce measured drawings using scale, plans, sections and elevations.
- Analyse a site for orientation, drainage, access, noise and climate.
- Use architectural vocabulary accurately when presenting or critiquing work.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

The design process

Design is not decoration applied at the end. It is a sequence of decisions from problem to built response.

- Brief: what the client needs, who will use the building, what constraints exist.
- Site analysis: orientation, access, topography, vegetation, noise, views and services.
- Concept: the organising idea - a diagram, a spatial strategy, a response to climate or culture.
- Schematic design: plans, sections and massing at small scale.
- Design development: materials, structure, services and detail coordination.
- Presentation: drawings, physical or digital models, and a clear verbal explanation.

Concept is not decoration
A concept explains why the building is organised the way it is. 'I used curves because they are beautiful' is a preference, not a concept.

Drawing and scale

Architectural drawing is a precise language. Learn the conventions and your work becomes legible to every reviewer.

- Plan: a horizontal cut about 1.2 metres above floor level.
- Section: a vertical cut showing heights and construction.
- Elevation: an external face of the building, without perspective.
- Scale: 1:100 or 1:50 for design drawings; 1:20 or 1:5 for details.
- Line weights: thin for construction lines, heavy for cut elements, medium for visible edges.
- Always include a north arrow, scale bar, title block and human figure for scale.

Site, climate and comfort

In a hot-humid climate such as Awka, passive design is not optional. Orientation and shading decide whether a room is usable.

- Long axis east-west reduces direct sun on long facades.
- Deep verandahs, brise-soleil and shaded openings reduce heat gain.
- Cross ventilation needs inlet and outlet openings on opposite sides.
- Consider prevailing winds, rainfall intensity and drainage before placing openings.
- Access and circulation: separate pedestrian and vehicle entry, and respect setbacks.
- Check the site's slope; it determines plinth levels and drainage strategy.

Critique language
Say 'the circulation is legible' or 'the service core is poorly located'. Avoid 'nice' and 'ugly'. Describe the effect, then judge it.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. What is the difference between a plan and an elevation?

Answer: A plan is a horizontal section cut through the building showing spaces and walls from above. An elevation is an orthographic view of an external face showing heights without perspective.

2. Why is a north arrow essential on a drawing?

Answer: It shows orientation, which determines solar exposure, prevailing wind and daylight strategy.

3. Give two passive strategies for a hot-humid climate.

Answer: Cross ventilation and shading with wide roof overhangs or brise-soleil. Others include light-coloured roofs and raised floors.

4. What should a site analysis cover?

Answer: Access, topography, orientation, climate, vegetation, drainage, noise, views, neighbouring buildings and available services.

5. What is a concept in architecture?

Answer: The organising idea that explains the design decisions and how the building responds to brief, site and use.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Measure your room and draw it to 1:50 with correct line weights.
- [ ] Draw the plan, section and one elevation of a simple building.
- [ ] Analyse your hostel site for sun, wind, access and drainage.
- [ ] Collect ten architectural terms and write a one-line definition for each.
- [ ] Sketch three concepts for a small campus kiosk with written justification.
- [ ] Visit one building on campus and critique its circulation in 150 words.
- [ ] Ask your studio tutor to review your drawing conventions before submission.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date: